From Stochasticity to Stability: A Monte Carlo-Guided Generative Prior Paradigm for Remote Sensing Change Detection

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***Abstract*—While denoising diffusion probabilistic model (DDPM) has demonstrated powerful capabilities in modeling the latent semantic distributions of remote sensing (RS) scenes, its application to change detection (CD) is severely limited by two inherent limitations. On the one hand, the inherent stochasticity of the diffusion process leads to representational instability when the model extracts internal features of changing regions. On the other hand, the DDPM's emphasis on global distribution modeling tends to limit its sensitivity to subtle local details, resulting in blurred change boundaries and degraded localization accuracy. To address these issues, we propose a Monte Carlo-guided generative prior paradigm from stochasticity to stability for CD, named S2S-MGPNet. First, we introduce DDPM as a feature extractor, leveraging its learned robust prior distribution to capture key semantic representations in complex RS scenes. Second, to address the representation instability induced by the inherent stochasticity of DDPM, we propose a Monte Carlo-guided spatiotemporal stabilization (MCST) strategy. Inspired by Monte Carlo estimation, this strategy aggregates multistep diffusion features and performs cross-scale spatial calibration to transform stochastic feature responses into stable representations, thereby enhancing feature stability within changed regions. Finally, to decode and refine the differential features, a cross-layer cascaded refinement (CLCR) strategy is incorporated to recover high-frequency details and reconstruct precise change boundaries. Experimental results demonstrate significant performance improvements across three public datasets. Specifically, compared with the traditional discriminative feature extraction paradigm, our method increased the IoU by 4.25% and 2.57% on the CDD and WHU datasets, respectively. The code will be publicly available at https://github.com/liaoyy-330/S2S-MGPNet**

***Index Terms*—Change detection (CD), diffusion model, remote sensing (RS) image, semantic feature extraction, stochasticity.**

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I. Introduction

REMOTE sensing image change detection (RSCD) utilizes multitemporal Earth observation data to identify surface changes within a specific geographical area [1]. This technique plays a key role in diverse applications, including urban planning [2], land use mapping [3], environmental monitoring [4], and disaster assessment [5].

Recently, the rapid evolution of deep learning has facilitated the adoption of deep neural networks, such as autoencoders [6], convolutional neural networks (CNNs) [7], and vision transformers (ViTs) [8] in RSCD. Leveraging robust feature learning capabilities, these models automatically extract abstract features ranging from low-level textures to high-level semantics directly from remote sensing (RS) images, thereby significantly enhancing detection efficiency and accuracy. However, the rapid development of spaceborne sensors has led to the widespread availability of high- and super-resolution images, thereby causing new challenges for RSCD. Traditional methods, which primarily focus on pixel-level spectral variations, often lack sufficient spatial structural modeling capabilities and consequently fail to capture the complex multiscale features and geometries inherent in RS imagery. CNN-based change detection (CD) methods [9], [10], [11] exhibit strong local feature extraction capabilities, effectively capturing fine-grained details such as textures and edges. However, their inherently local receptive fields limit the ability to capture long-range spatial dependencies, resulting in insufficient modeling of global contextual information. To address this limitation, Transformer-based networks [12], [13], [14] have been introduced to capture long-range spatial dependencies through self-attention mechanisms. Nevertheless, the

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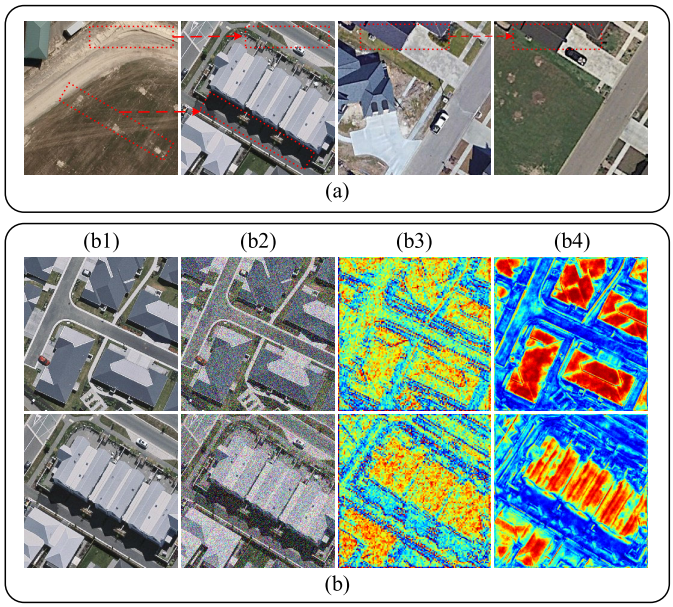


Fig. 1. (a) Illustration of pseudo-changes in complex RS scenes. (b) Visualization of distributed diffusion features enhanced by the MCST strategy, where (b1) denotes the original image, (b2) denotes the image input into the CD model after forward noise addition, (b3) denotes the feature heatmaps extracted by DDPM, and (b4) denotes the enhanced feature heatmaps processed by MCST.

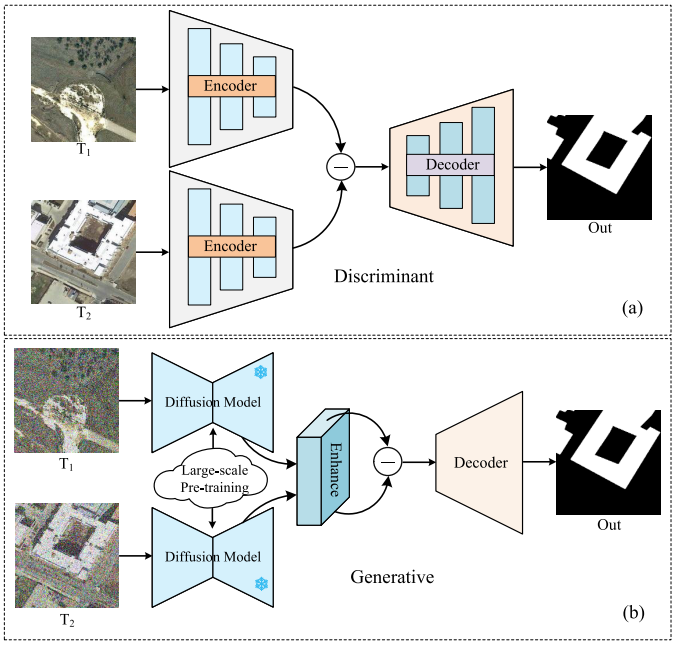


Fig. 2. Comparison between traditional discriminative paradigms and generative S2S-MGPNet. (a) Traditional discriminative network framework: Directly learns mappings from input to output. (b) Generative network framework: By pretraining on massive RS imagery, the DDPM learns the intrinsic distribution of the data. Subsequently, the DDPM is frozen and utilized as a feature extractor. Since the DDPM is fundamentally based on a UNet architecture, the output from each layer of its decoder is extracted to serve as the feature input for the CD task.

inherent class imbalance in CD often biases global attention weights heavily toward unchanged backgrounds, thereby limiting its ability to effectively focus on sparse, localized change signals. Furthermore, imaging inconsistencies in RS data (e.g., seasonal changes or lighting conditions) often induce pseudo-changes [see Fig. 1(a)], which hinder traditional discriminative CD paradigms [shown in Fig. 2(a)] from accurately discerning genuine semantic change features.

As a powerful class of generative models, diffusion models present a novel feature extraction paradigm for RSCD. Their primary advantage lies in the capacity to learn intrinsic data distributions and essential characteristics through a progressive denoising process that starts from pure noise. Due to the complex spatial distribution of ground objects, imaging noise, and pseudo-changes induced by seasonal variations in RS imagery, traditional discriminative models often struggle to effectively represent genuine change features. The denoising diffusion probabilistic model (DDPM), however, can model the latent data distribution through a stochastic denoising process, thereby capturing structural information in complex scenes more effectively. Furthermore, the generative nature of DDPM can provide an effective prior distribution for RS images, which is crucial for enhancing the model's robustness against noise interference. Consequently, incorporating diffusion model-based generative priors into the CD framework presents a highly promising research direction for enhancing feature representation capabilities. However, DDPM faces two key challenges in feature extraction: First, the semantic information extracted by DDPM exhibits strong stochasticity [15], [16], making it difficult to generate stable target object features. Second, the emphasis of DDPM on global distribution means it tends to neglect local details, causing blurred change boundaries and insufficient localization accuracy.

To overcome the aforementioned challenges, we propose S2S-MGPNet (a diffusion-based CD framework), whose overall generative architecture is illustrated in Fig. 2(b). First, we introduce a pretrained DDPM [16] as a feature extractor, leveraging the prior distribution learned during its denoising process to capture robust semantic features, as visualized in Fig. 1(b3). Second, we design the Monte Carlo-guided spatiotemporal stabilization (MCST) strategy, which adopts a Monte Carlo ensemble (MCFA)-based strategy in the temporal dimension to suppress the stochasticity of the diffusion process, and integrates multiscale feature fusion in the spatial domain to enhance feature consistency within change regions. The enhanced features are shown in Fig. 1(b4). Finally, we propose a cross-layer cascaded refinement (CLCR) strategy, which reconstructs high-frequency spatial information through cross-layer feature reuse and parallel attention mechanisms, thereby alleviating boundary blurring. The main contributions of this article can be summarized as follows.

1) *Generative Prior-Based Paradigm:* Unlike traditional discriminative paradigms that are limited to fitting rigid decision boundaries on finite training samples while neglecting the intrinsic distribution of RS data, which often leads to poor generalization in complex RS scenes, we design an S2S-MGPNet. It adopts a distributed feature extraction framework based on generative priors, which effectively resolves the semantic representation bottleneck of discriminative paradigms by leveraging

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the strong distributed semantic perception capability of pretrained DDPM.

2) *Spatiotemporal Joint Representation:* Unlike existing diffusion-based methods that focus on spatial feature fusion while overlooking the inherent stochasticity within the diffusion process, which leads to representational instability, we propose an MCST strategy. By designing a joint modeling mechanism across both temporal and spatial dimensions, we effectively mitigate the inherent stochasticity and achieve robust feature representation for complex RS scenes.
3) *Cross-Layer Boundary Perception:* Unlike existing methods that rely on single-scale feature decoding while neglecting the loss of high-frequency details in deep layers, resulting in blurred boundaries, we propose a CLCR strategy. By designing a bottom-up cascading recovery pathway, this strategy effectively resolves the issue of boundary fuzziness and reconstructs fine-grained structural details from coarse representations.

The remainder of this article is organized as follows. Section II reviews the applications of traditional discriminative and generative models in CD, as well as discriminative multiscale feature fusion methods. Section III elaborates on the overall framework and core modules of the proposed method. Section IV presents comparative experiments with state-of-the-art methods and comprehensive ablation studies. Finally, Section V concludes this article.

II. Related Works

A. Discriminative Models for RSCD

In recent years, discriminative deep learning models represented by CNNs and Transformers have become the mainstream paradigm in the RSCD field due to their powerful end-to-end learning capabilities. Based on their core architectural characteristics, existing methods can be broadly categorized into two types.

1) CNN-Based Methods: SNUNet [17], UNet++ [18], and A2Net [19] typically adopt the classic Siamese Network architecture. In such architectures, a pair of weight-shared encoders is employed to independently extract features from bitemporal images, followed by computing difference features to identify change regions. In early research, Daudt et al. [20] proposed two fully convolutional Siamese extensions based on the UNet backbone: FC-Siam-conc employs feature concatenation for fusion, whereas FC-Siam-diff utilizes feature subtraction to obtain change features. Subsequently, Guo et al. [21] built upon this foundation by integrating the Siamese network with a contrastive loss to mitigate the impact of pseudo-changes, thereby improving detection accuracy. Furthermore, Chen et al. [22] combined CNNs with LSTM networks to enhance bitemporal feature correlation via temporal modeling. To further enhance the capability of capturing feature differences, Feng et al. [23] adopted a dual-branch architecture that combines pixel-wise subtraction and channel-wise concatenation, which not only enables intuitive modeling of feature differences but also compensates for the limitations of a single operation through learnable concatenation.

2) Transformer-Based Methods: To overcome the inherent limitations of CNNs in modeling long-range dependencies, ViTs and their variants have been introduced into the RSCD domain due to their strong global context modeling capabilities. For instance, Zhang et al. [24] constructed a Siamese-CD network by stacking multiple Swin Transformer blocks to effectively capture global contextual information. To fully exploit bi-temporal global context information, Yan et al. [12] utilized the Swin Transformer to learn discriminative global features and adopted a feature pyramid structure to aggregate multilayer features, thereby enhancing the model's representational capacity. However, individual CNNs or Transformer architectures struggle to simultaneously model both local details and global context efficiently. Furthermore, the computational complexity of self-attention in Transformers scales quadratically. To address these challenges, Chen et al. [25] proposed a CNN-Transformer hybrid architecture, where CNNs extract local features and Transformers model global dependencies. Building upon this paradigm, Xu et al. [14] further advanced CNN-Transformer integration by adopting an interleaved fusion architecture, which enables more frequent local–global interactions while significantly reducing computational overhead and maintaining high performance.

B. Generative Models for RSCD

Recently, DDPM has demonstrated remarkable potential in tasks such as image generation [26], [27], [28], reconstruction [29], [30], [31], and segmentation [32], [33], [34], owing to their strong capabilities in learning data distributions. This has opened new avenues for RSCD. In contrast to traditional discriminative models, DDPM captures the nonlinear feature distributions of complex RS scenes via a progressive denoising process, thereby enabling more precise identification of authentic semantic features. Furthermore, DDPM implicitly models global context during the generative process, effectively overcoming the local receptive field limitations of CNNs while mitigating the quadratic computational complexity associated with Transformers. Within the RSCD domain, existing diffusion-based approaches can be broadly categorized into two technical paradigms according to their utilization of diffusion models:

1) Feature Extractor: The first category treats DDPM as a powerful feature extractor to enhance the semantic representation of bi-temporal RS imagery for CD [16], [35], [36], [37]. For instance, Zhang et al. [37] pioneered the application of a semantic correlation diffusion model (SCDM) for hyperspectral image CD. SCDM effectively leverages unlabeled data to extract spectral–spatial semantic correlation features, thereby mitigating the impact of pseudo-changes. Similarly, Bandara et al. [16] pretrained a DDPM on unlabeled data and subsequently employed it to extract feature representations from an RS image. This approach generated context-rich semantic features to augment the downstream CD network, achieving promising results.

2) Conditional Mask Generation: The second category formulates CD as a conditional mask generation task [38], [39], [40], [41], treating DDPM as a generative framework to progressively synthesize change masks. For instance,

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Wen et al. [40] proposed GCD-DDPM, a model that directly generates CD results via DDPM. By adaptively recalibrating predictions during inference, it accurately identifies subtle and irregular changes in complex scenes. Furthermore, Zhao et al. [41] introduced DDPM into multimodal RSCD tasks. By employing a bidirectional diffusion mechanism to bridge multimodal discrepancies, they achieved collaborative CD across arbitrary RS modalities.

C. Discriminative Multiscale Feature Fusion Strategy

Multiscale feature fusion is designed to capture both spatial details and abstract semantics across hierarchical levels, thereby providing highly discriminative feature representations for change regions. Accordingly, existing studies can be broadly categorized into three primary dimensions: spatial structure alignment, bitemporal feature correction, and difference information refinement.

1) Spatial Structure Alignment: This category focuses on addressing spatial information loss and feature misalignment induced by the downsampling operations in deep feature extraction. To tackle this, Li et al. [42] proposed a flow-guided upsampling module, which reduces information loss and rectifies spatial bias by aligning feature maps across different resolutions. Similarly, Feng et al. [43] designed a cross-scale feature interaction module. By facilitating feature flow and interaction across hierarchical layers, this method effectively circumvents the target spatial misalignment caused by backbone downsampling, thereby enhancing detection accuracy for small objects. Furthermore, Shangguan et al. [44] developed a multiscale attention fusion encoder. By leveraging attention mechanisms to adaptively weight and fuse adjacent-scale features, this approach effectively restores shallow spatial structures and edge details while preserving deep spectral semantics.

2) Bitemporal Feature Correction: This category aims to reduce pseudo-changes caused by imaging perspective differences or registration errors. To address this, Liu and Shi [45] introduced deformable convolutional networks (DCNs), which align bi-temporal features by dynamically adjusting the sampling positions of convolutional kernels. This approach facilitates the adaptive correction of geometric deformations at the feature level, significantly mitigating the adverse impact of registration errors on CD results.

3) Difference Feature Refinement: This category focuses on optimizing difference feature fusion strategies to enhance the saliency of difference features. Moving beyond simple feature subtraction, Li et al. [19] designed a progressive difference feature identification module. This module extracts and amplifies difference information from fused features in a coarse-to-fine manner, thereby enabling the precise capture of subtle change regions.

Beyond CD tasks, recent advances in general RS multiscale feature decoupling and dynamic fusion also provide a solid foundation for complex scene perception. For instance, Wu et al. [46] achieve the spatial decoupling of contours and textures, while Xie et al. [47] implement boundary-guided dynamic feature enhancement. In complex dense scenes, Yu and Xu [48] accomplish bidirectional multiscale adaptive fusion. Furthermore, Sun et al. [49] realize dynamic feature extraction and redundancy decoupling in high-dimensional spectral spaces. The design principles from these works provide valuable insights for the hierarchical feature refinement and spatial calibration within our S2S-MGPNet.

Although multiscale feature fusion strategies have been extensively adopted in the RSCD domain, existing methods are predominantly built upon discriminative networks, leaving a significant gap in research regarding fusion strategies for the distributed, stochastic features extracted by DDPM. To bridge this gap, this article designs a fusion strategy tailored for distributed diffusion features. Specifically, we employ an MCFA-based stabilization mechanism to suppress the inherent stochasticity of the diffusion process. Subsequently, inspired by multiscale feature decoupling, we leverage the inherent hierarchical correlations of diffusion features to aggregate adjacent-level information. This enhances current-layer representations, thereby achieving cross-scale feature calibration to establish deterministic representations.

III. Method

In this section, we first introduce the preliminaries of DDPM. Subsequently, the overall architecture of the proposed S2S-MGPNet framework is presented. The details of the MCST and CLCR strategies are then elaborated upon. Finally, the objective loss function is formulated.

A. Preliminaries of DDPM

The DDPM is a class of generative models grounded in Markov chains. It consists of two distinct processes: the forward diffusion process and the reverse denoising process.

The forward diffusion process aims to progressively perturb the original data $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ by adding Gaussian noise until it converges to an isotropic standard Gaussian distribution $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Given a predefined variance schedule $\beta_1, \beta_2, \ldots, \beta_T$, where $\beta_t \in (0, 1)$, the state $\mathbf{x}_t$ at any time step t depends solely on the previous state $\mathbf{x}_{t-1}$. The transition kernel is defined as

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}\right) \tag{1}$$

where T denotes the total number of time steps. Leveraging the properties of Gaussian distributions and the reparameterization trick, we can directly sample the noisy image $\mathbf{x}_t$ at any arbitrary time step t from $\mathbf{x}_0$ without iterative stepping. Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$. $\mathcal{N}$ denotes a Gaussian probability density function. The marginal distribution $q(\mathbf{x}_t|\mathbf{x}_0)$ can be expressed as

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t;\ \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}\right). \tag{2}$$

Specifically, $\mathbf{x}_t$ can be explicitly calculated as follows:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \tag{3}$$

This property enables efficient generation of samples with varying noise intensities during the training phase, laying the foundation for subsequent feature extraction and denoising tasks.

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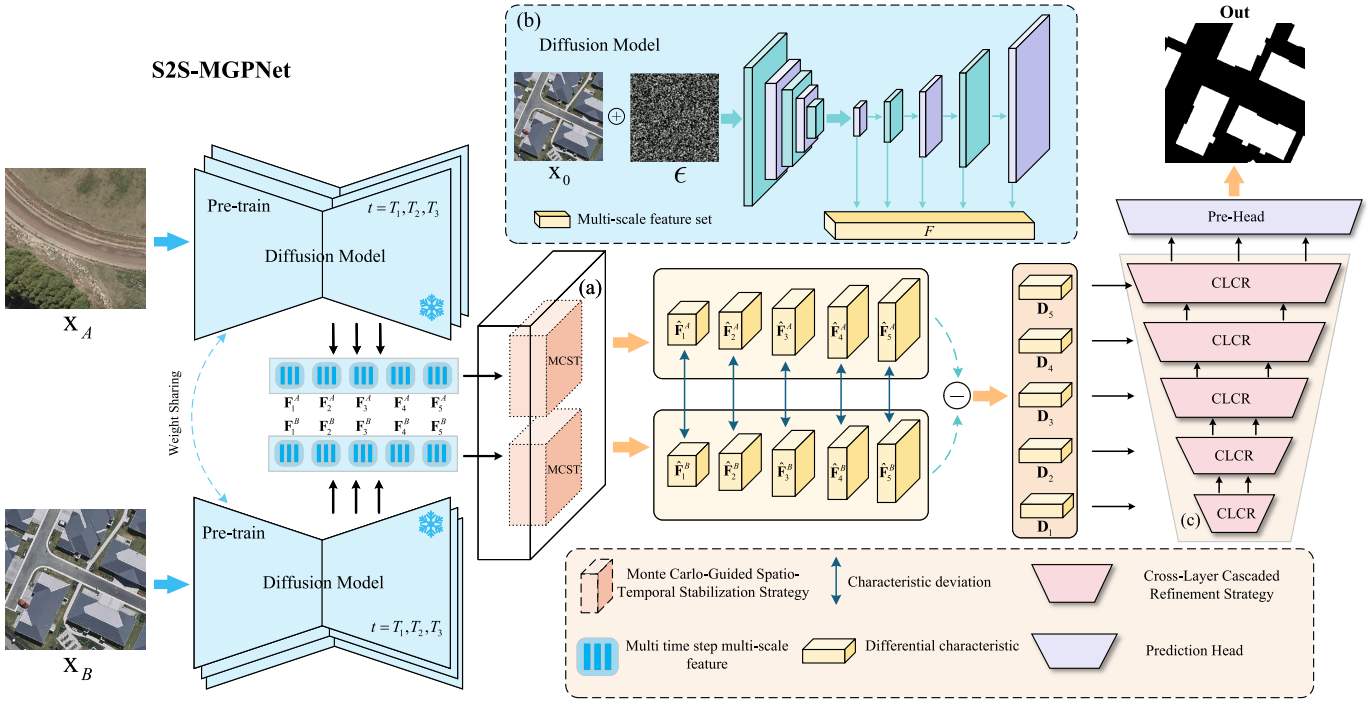


Fig. 3. Overall architecture of the proposed S2S-MGPNet. The areas with a blue background denote the frozen parameters, while the remaining colors represent the trainable parameters for the CD task. Stage 1: Bi-temporal images $(\mathbf{x}_A^t, \mathbf{x}_B^t)$ are perturbed with noise corresponding to the time step and fed into the DDPM. The outputs from the DDPM's each decoder layer serve as multiscale features. Stage 2: We employ the MCST strategy to enhance the extracted bi-temporal distributed diffusion feature sets $(\mathbf{F}_n^A, \mathbf{F}_n^B)$, effectively achieving the transformation from stochastic features to stable representations. Subsequently, feature subtraction is performed to obtain difference features at corresponding scales. Finally, the CLCR strategy is employed to decode and predict the final CD result. (a) MCST feature enhancement strategy for enhancing generative features, (b) pre-trained DDPM for extracting generative features, and (c) denotes the difference feature decoder, which applies the CLCR strategy in each layer for feature decoding.

B. Overall Framework

Currently, preliminary progress has been made in applying DDPM to RSCD tasks. However, to achieve stable feature extraction, fully exploit global information, and enhance local details, further refinement of the distributional and stochastic characteristics of DDPM outputs is necessary. To address these challenges, this article proposes an efficient DDPM-assisted RSCD model, with the overall architecture illustrated in Fig. 3. This work aims to bridge the gap in existing research concerning the processing and fusion of distributed diffusion features, thereby fully unlocking the potential of DDPM for RSCD. The proposed workflow comprises two stages: Stage 1 focuses on acquiring semantic features for the RSCD task. Stage 2 is dedicated to feature enhancement and difference feature extraction, ultimately generating a binary CD result via upsampling.

1) Feature Extraction: To enhance the model's generalization ability to complex RS scenes and effectively capture potential semantic distributions, this article adopts a feature extraction strategy based on DDPM, which is essentially a UNet-based noise predictor as illustrated in Fig. 3(b). The UNet is used to predict the added Gaussian noise, and the model is optimized by minimizing the difference between the predicted and ground-truth noise, thus enabling the UNet to have powerful simulation capabilities for complex data distributions. To enable DDPM to accurately locate RS semantic features, we conducted large-scale unsupervised pretraining before feature extraction. Specifically, the model was trained using RS images collected by Google Earth Engine [50] from approximately 200 K locations worldwide, enabling the model to learn robust semantic representations of RS features. We use the pretrained and frozen DDPM to extract distributed diffusion features.

To improve model generalization and adaptability to noise, we perturb the input bitemporal images $\mathbf{x}_A$ and $\mathbf{x}_B$ with Gaussian noise at time step t, as formulated in (4). This process generates the noisy bitemporal pair $\hat{\mathbf{x}}_A^t$ and $\hat{\mathbf{x}}_B^t$

$$\begin{cases} \hat{\mathbf{x}}_A^t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_A + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon} \\ \hat{\mathbf{x}}_B^t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_B + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon} \end{cases} \tag{4}$$

where a larger time step t corresponds to a higher noise intensity, and $\bar{\alpha}_t$ denotes the cumulative product of the hyperparameter α_t from 1 to t, governing the magnitude of the injected noise.

Subsequently, $\hat{\mathbf{x}}_A^t$ and $\hat{\mathbf{x}}_B^t$ are input into the DDPM, where the output of each layer of the DDPM decoder is extracted as the multiscale feature corresponding to the current noise intensity, as follows:

$$\begin{cases} \left(\mathbf{F}_{t,1}^A, \mathbf{F}_{t,2}^A, \mathbf{F}_{t,3}^A, \mathbf{F}_{t,4}^A, \mathbf{F}_{t,5}^A\right) = \phi_{\text{DDPM}}\left(\hat{\mathbf{x}}_A^t\right) \\ \left(\mathbf{F}_{t,1}^B, \mathbf{F}_{t,2}^B, \mathbf{F}_{t,3}^B, \mathbf{F}_{t,4}^B, \mathbf{F}_{t,5}^B\right) = \phi_{\text{DDPM}}\left(\hat{\mathbf{x}}_B^t\right) \end{cases} \tag{5}$$

where $\phi_{\text{DDPM}}(\cdot)$ denotes the pretrained DDPM, $\mathbf{F}_{t,n}^A$ represents the hierarchical features, A and B denote the two temporal phases, n denotes the specific network layer, and $t \in T_n$ represents the specific time step.

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Because the DDPM captures different feature information at various diffusion stages, we perturb the original images at three distinct time steps. In the low-noise interval ($T_1 = 50$), the signal-to-noise ratio is high, and the model primarily captures high-frequency spatial structures like building boundaries and road textures. The medium-noise interval ($T_2 = 150$) is a transition where signal and noise are balanced; for the RSCD task, these features help bridge the gap between abstract semantics and pixel-level textures, ensuring the consistency of changed objects. In the high-noise interval ($T_3 = 250$), pixel-level details are corrupted, forcing the model to focus on global semantic distributions, which helps suppress high-frequency "pseudo-changes" (e.g., sensor noise and illumination fluctuations). By independently feeding the perturbed images into the DDPM, we extract the corresponding multiscale feature sets as follows:

$$\begin{cases} \mathbf{F}_n^A = \left[\mathbf{F}_{t_1,n}^A, \mathbf{F}_{t_2,n}^A, \mathbf{F}_{t_3,n}^A\right], & n \in \mathbb{N}^+, 1 \le n \le 5 \\ \mathbf{F}_n^B = \left[\mathbf{F}_{t_1,n}^B, \mathbf{F}_{t_2,n}^B, \mathbf{F}_{t_3,n}^B\right], & n \in \mathbb{N}^+, 1 \le n \le 5 \end{cases} \tag{6}$$

where n denotes different layers, and $\mathbf{F}_n^A$ represents the multi-time-step feature set at the nth layers in phase A.

2) Change Detection: Most existing CD methods predominantly rely on discriminative networks, which often struggle to adequately capture correlations among local details. Furthermore, there is a lack of CD strategies for processing distributed diffusion features. Distinct from existing methods, an MCST strategy is designed. It employs a Monte Carlo strategy to aggregate multi-time-step features for suppressing diffusion stochasticity. Subsequently, through complementary neighborhood spatial information, the representation capability for complex regions is enhanced. Specifically, the extracted bi-temporal multiscale distributed diffusion feature sets $(\mathbf{F}_n^A, \mathbf{F}_n^B)$ are enhanced by the MCST strategy, yielding the strengthened feature sets $(\hat{\mathrm{F}}_n^A, \hat{\mathrm{F}}_n^B)$. Subsequently, the multilayer difference features $\mathbf{D}_n$ are obtained, as formulated in the following equations:

$$\left(\hat{\mathrm{F}}_n^A, \hat{\mathrm{F}}_n^B\right) = \mathcal{M}_{\mathrm{MCST}}\left(\mathbf{F}_n^A, \mathbf{F}_n^B\right) \tag{7}$$

$$\mathbf{D}_n = \left|\hat{\mathrm{F}}_n^A - \hat{\mathrm{F}}_n^B\right|, \quad n \in \mathbb{N}^+, 1 \le n \le 5 \tag{8}$$

where $\mathcal{M}_{\mathrm{MCST}}(\cdot)$ denotes the MCST strategy, $|\cdot|$ denotes the pixel-wise difference calculation.

Differential features generated through direct pixel-wise subtraction are typically coarse and lack semantic robustness. Furthermore, the inherent emphasis of DDPM on global distributions leads to the loss of fine-grained change details, causing blurred change boundaries. To address this issue, a strategy of CLCR is designed to enhance and fuse multiscale differential features, as follows:

$$\hat{\mathrm{D}}_n = \begin{cases} \mathcal{F}_{\mathrm{bound}}\left(\mathbf{D}_n\right), & \text{if } n = 1 \\ \mathcal{F}_{\mathrm{bound}}\left(\hat{\mathrm{D}}_{n-1} + \mathbf{D}_n\right), & \text{otherwise} \end{cases} \tag{9}$$

where $\mathcal{F}_{\mathrm{bound}}(\cdot)$ denotes CLCR, and $\hat{\mathrm{D}}_n$ represents the enhanced differential feature.

Finally, the hierarchically enhanced $\hat{\mathrm{D}}_n$ is fed into the prediction head to predict the final CD result map, as follows:

$$\boldsymbol{Out} = \mathrm{Conv}_{3\times3}\left(\mathrm{Conv}_{3\times3}\left(\hat{\mathrm{D}}_5\right)\right) \tag{10}$$

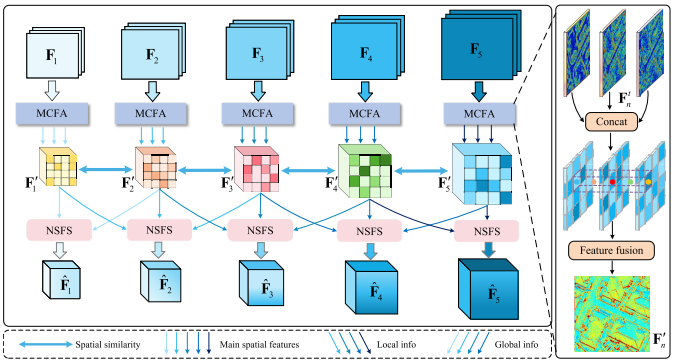


Fig. 4. MCST strategy, where MCFA is a Monte Carlo-based feature aggregation strategy, and NSFS is an NSFS for airspace.

where $\boldsymbol{Out}$ denotes the CD result, and $\mathrm{Conv}_{3\times3}(\cdot)$ denotes the 3×3 convolution. Specific details of MCST and CLCR will be elaborated in Sections III-B and III-C.

C. MCST Strategy

Unlike traditional discriminative networks that produce deterministic feature responses, the inherent stochasticity of DDPM induces fluctuations in the numerical distribution of features extracted at a single time step. To achieve stable modeling of complex RS scenes, this article proposes an MCST strategy. As illustrated in Fig. 4, MCST aims to construct robust and stable feature representations through a feature enhancement strategy in both temporal and spatial dimensions.

1) In the Temporal Dimension: Effectively suppressing the stochastic noise in DDPM is crucial for extracting stable semantic features from RS imagery. However, even given fixed $\mathbf{x}_A$ and t, the features extracted by DDPM remain random variables. This stochasticity stems from the random noise $\boldsymbol{\epsilon}$ injected during the diffusion process, as shown in (11). Feature extraction at a single time step is equivalent to a point estimation of this distribution, which implies high statistical variance and manifests as feature instability

$$\begin{cases} \mathbf{F}_n^A = \phi_{\mathrm{DDPM}}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_A + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\right) \\ \mathbf{F}_n^B = \phi_{\mathrm{DDPM}}\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_B + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}\right) \end{cases} \tag{11}$$

where $\mathbf{x}_A$ and $\mathbf{x}_B$ represent the original images, t denotes the time step, $\bar{\alpha}_t$ is the preset noise coefficient, and $\boldsymbol{\epsilon} \in \mathcal{N}(\mathbf{0}, \mathbf{I})$ is a Gaussian noise from a standard normal distribution.

To obtain a stable feature representation $\mathbf{F}_n'$ of original RS image, we employ a feature stabilization strategy based on MCFAs. Specifically, we independently sample three noise signals at three distinct time steps $t \in \{T_1, T_2, T_3\}$ to perform independent forward noise injection and feature extraction. This yields a temporal feature set $\mathbf{F}_n = \{\mathbf{F}_n^{T_1}, \mathbf{F}_n^{T_2}, \mathbf{F}_n^{T_3}\}$ at each scale. Consequently, the feature is no longer a fragile single-point estimate but a small ensemble of feature responses to original images under varying noise perturbations. This feature set captures richer and more stable information than a single feature map.

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Traditional Monte Carlo methods typically employ arithmetic averaging to approximate expected values, as follows:

$$\mathbf{F}'_n = \frac{1}{N}\sum_{i=1}^{N}\mathbf{F}_n^{T_i} \tag{12}$$

where N is the number of time steps, and $\mathbf{F}_n^{T_i}$ denotes the feature at scale n perturbed at time step T_i.

For each independent time step, the extracted diffusion feature can be modeled as

$$\mathbf{F}_n^{T_i} = \mathbf{F} + \boldsymbol{\epsilon}_{T_i} \tag{13}$$

where $\mathbf{F}$ represents the stable semantic component, and $\boldsymbol{\epsilon}_{T_i} \sim \mathcal{N}(0, \sigma_{T_i}^2 I)$ denotes the noise introduced by random Gaussian sampling during the diffusion process, with a mean of 0 and a variance $\sigma_{T_i}^2$ dependent on the time step. The variance $\sigma_{T_i}^2$ increases as the noise level at T_i rises, indicating that the features at different time steps follow a heteroscedastic distribution. If an arithmetic mean operation is adopted, the variance of the enhanced feature $\mathbf{F}'_\mathbf{n}$ can be expressed as

$$Var_{avg}\left(\mathbf{F}'_n\right) = \frac{1}{N^2}\sum_{i=1}^{N}\boldsymbol{\epsilon}_{T_i} = \frac{1}{N^2}\sum_{i=1}^{N}\sigma_{T_i}^2. \tag{14}$$

According to the Gauss-Markov theorem, for homoscedastic noise, the traditional arithmetic mean yields the minimum variance unbiased estimator (MVUE). Conversely, for heteroscedastic samples, the MVUE is achieved by a weighted average inversely proportional to the sample variance. Moreover, simple arithmetic averaging often eliminates high-frequency details specific to different noise levels. To address these issues, we adopt a learnable nonlinear aggregation paradigm that adaptively learns the optimal weights corresponding to the noise variance at each time step. Consequently, it achieves a lower estimation variance than the arithmetic mean, as follows:

$$Var_{learn}\left(\mathbf{F}'_n\right) = \sum_{i=1}^{N}\boldsymbol{w}_i^2\sigma_{T_i}^2 \le Var_{avg}\left(\mathbf{F}'_n\right) = \frac{1}{N^2}\sum_{i=1}^{N}\sigma_{T_i}^2 \tag{15}$$

where $\boldsymbol{w}_i \in [0,1]$ denotes the normalized weight to the features at different time steps. The equality holds only when all samples are homoscedastic. For the heteroscedastic diffusion features in this article, the learnable aggregation can achieve a smaller estimation variance, which directly enhances the stability of the feature representation.

As illustrated in Fig. 4, this method utilizes a channel concatenation technique to ensure that the original information entropy is preserved across all time steps. Subsequently, learnable convolution kernels are employed to automatically capture the key features present under different noise intensities, thereby achieving weighted statistical aggregation. The specific process is as follows:

$$\mathbf{F}_{cat} = \mathrm{Conv}_{1\times1}\left(\mathrm{Cat}\left(\mathbf{F}_n^{T_1}, \mathbf{F}_n^{T_2}, \mathbf{F}_n^{T_3}\right)\right) \tag{16}$$

where $\mathrm{Conv}_{1\times1}(\cdot)$ denotes the 1×1 convolution, and $\mathrm{Cat}(\cdot)$ represents the channel concatenation operation. Expressing this operation in the form of a weighted sum yields

$$\mathbf{F}_{cat} = \sum_{i=1}^{T}\boldsymbol{w}_i \cdot \mathbf{F}_n^{T_i} + \mathbf{b}_c \tag{17}$$

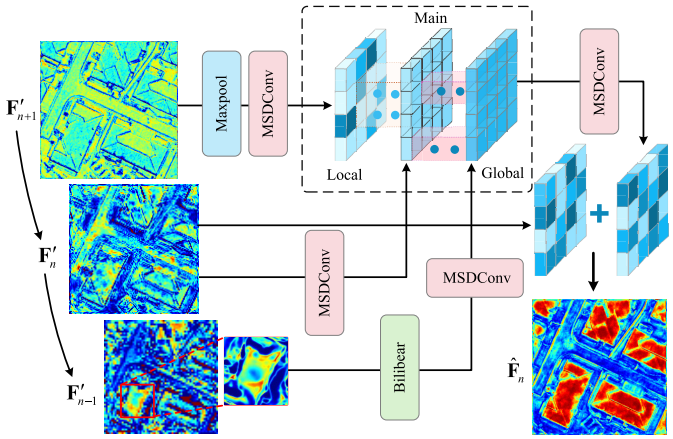


Fig. 5. NSFS. "Main" denotes the primary spatial information, "global" represents the global semantic calibration branch, and "local" refers to the local detail refinement branch. Residual learning is employed to preserve the primary stage information.

where $\boldsymbol{w}_i$ denotes the fusion weight for the feature at the corresponding time step, and $\mathbf{b}_c$ represents the bias term. During training, a weight normalization constraint is imposed on the 1×1 convolution, i.e., $\sum_{i=1}^{T}\boldsymbol{w}_i = 1$, and the bias term is set to 0. Under this constraint, the expectation of the output is as follows:

$$\mathbb{E}\left[\mathbf{F}_{cat}\right] = \sum_{i=1}^{T}\boldsymbol{w}_i \cdot \mathbb{E}\left[\mathbf{F}_n^{T_i}\right] = \mathbf{F} \tag{18}$$

where $\mathbb{E}[\cdot]$ denotes the mathematical expectation, and $\mathbf{F}$ represents the stable semantics. This proves that the 1×1 convolution is an unbiased estimator of $\mathbf{F}$. As T increases, this weighted estimator converges to $\mathbf{F}$, thereby ensuring the convergence of the feature distribution. Subsequently, a 3×3 convolution is applied to perform spatial weighting on the feature maps. This operation is equivalent to executing Monte Carlo averaging within a local spatial neighborhood, thereby achieving the weighted statistical integration of features, as follows:

$$\mathbf{F}'_n = \mathrm{Conv}_{3\times3}\left(\sigma\left(\mathbf{F}_{cat}\right)\right) \tag{19}$$

where $\mathrm{Conv}_{3\times3}(\cdot)$ denotes the 3×3 convolution, and $\sigma(\cdot)$ represents the ReLU activation function.

2) In the Spatial Dimension: Single-layer diffusion features confront the challenge of representational ambiguity, where deep features $\mathbf{F}'_{n-1}$ possess robust global semantics but lack geometric details, whereas shallow features $\mathbf{F}'_{n+1}$ preserve high-frequency textures yet contain substantial background noise. To address this, we design a neighborhood spatial multilayer feature fusion strategy (NSFS). This strategy effectively mitigates the representational ambiguity and uncertainty inherent in single-layer features, rendering the enhanced features spatially stable and precise. The structure of NSFS is illustrated in Fig. 5.

This strategy treats the current layer feature as an "anchor point" and leverages adjacent hierarchical features as auxiliary context, constructing two calibration branches to reinforce the "anchor point" feature representation.

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1) *Global Semantic Calibration Branch:* This branch regards the deep feature $\mathbf{F}'_{n-1}$ as a semantic guide, injecting deep-layer contextual information via upsampling to suppress semantic ambiguity at the current layer.
2) *Local Detail Refinement Branch:* This branch utilizes the shallow feature $\mathbf{F}'_{n+1}$ as a texture reference, introducing fine-grained boundary information by downsampling to rectify geometric deviations at the current layer.

This cross-scale information injection is equivalent to establishing a bidirectional spatial calibration field, enabling the features to possess both deep-layer semantic stability and shallow-layer detail precision. Specifically, to resolve scale mismatch between adjacent layers, we apply bilinear interpolation to upsample the deep feature $\mathbf{F}'_{n-1}$ and maxpooling to downsample the shallow feature $\mathbf{F}'_{n+1}$, aligning their resolutions with the current layer $\mathbf{F}'_n$, as follows:

$$\begin{cases} \tilde{\mathbf{F}}_{n+1} = \mathcal{D}_{\text{maxpool}}\left(\mathbf{F}'_{n+1}\right) \\ \tilde{\mathbf{F}}_{n} = \mathbf{F}'_{n} \\ \tilde{\mathbf{F}}_{n-1} = \mathcal{U}_{\text{bilinear}}\left(\mathbf{F}'_{n-1}\right) \end{cases} \tag{20}$$

where $\tilde{\mathbf{F}}$ denotes the aligned feature tensor, $\mathcal{D}_{\text{maxpool}}(\cdot)$ represents maxpooling, and $\mathcal{U}_{\text{bilinear}}(\cdot)$ denotes bilinear upsampling. Specifically, for the top-layer and bottom-layer features, we only utilize a single adjacent layer for alignment (as shown in Fig. 5, where the local branch and global branch are omitted for the top and bottom layers, respectively).

To fully exploit the aligned contextual information, we introduce the multiscale decoupled convolutional block (MSDConv) [51] to decouple and refine features for each branch, thereby achieving the hierarchical refinement of the diffusion features. MSDConv effectively mitigates the feature over-smoothing problem caused by standard convolutions. While naturally aligning with the multiscale hierarchical characteristics of the diffusion features, it successfully suppresses their inherent randomness, thereby significantly improving the accuracy of spatial fusion. Subsequently, the features are concatenated along the channel dimension to achieve multilayer feature fusion, as follows:

$$\mathbf{F}^n_{\text{cat}} = \left[\Psi_{\text{MSD}}\left(\tilde{\mathbf{F}}'_{n-1}\right), \Psi_{\text{MSD}}\left(\tilde{\mathbf{F}}'_{n}\right), \Psi_{\text{MSD}}\left(\tilde{\mathbf{F}}'_{n+1}\right)\right] \tag{21}$$

where $\Psi_{MSD}(\cdot)$ denotes the MSDConv convolution operation, and $[\cdot]$ represents concatenation along the channel dimension.

To ensure that adjacent-layer features serve solely as supplementary residuals for calibrating the current-layer feature, we employ a residual learning mechanism. Specifically, the aggregated feature $\mathbf{F}^n_{cat}$ first undergoes channel dimensionality reduction and information fusion via the MSDConv convolution. Subsequently, it is combined with the current-layer feature $\tilde{\mathbf{F}}'_n$ through pixel-wise weighted reconstruction to yield the final spatially enhanced feature $\hat{\mathrm{F}}_n$, as follows:

$$\hat{\mathrm{F}}_n = \text{Conv}_{1\times 1}\left(\tilde{\mathbf{F}}'_n\right) + \Psi_{\text{MSD}}\left(\mathbf{F}^n_{\text{cat}}\right). \tag{22}$$

Through the MCST strategy, enhanced bi-temporal distributed diffusion features $\{\hat{\mathrm{F}}^A_n, \hat{\mathrm{F}}^B_n\}^5_{n=1}$ are obtained. Subsequently, a pixel-wise absolute difference strategy is employed to acquire the multiscale difference features $\{\mathbf{D}_n\}^5_{n=1}$.

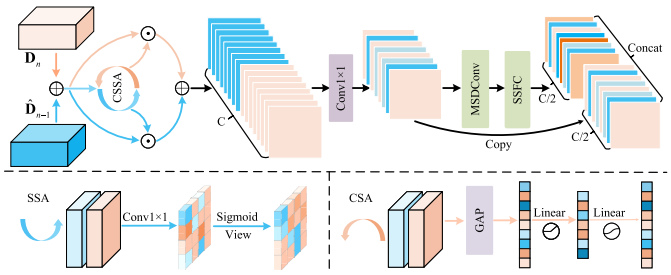


Fig. 6. CLCR. "CSSA" denotes the parallel spatial-channel attention mechanism, "SSA" denotes the spatial attention, "CSA" denotes the channel attention, and C denotes the number of channels. This mechanism enhances boundary information.

D. CLCR Strategy

Although the differential features obtained in the previous stage capture basic change information, they often suffer from boundary fragmentation and high-frequency detail loss due to the resolution degradation in deep layers. To accurately reconstruct the lost high-frequency boundary information, the CLCR strategy is designed. Distinct from traditional methods that rely on processing single-layer features, CLCR constructs a bottom-up decoding path. It leverages the rich spatial details contained in shallow features to serve as guidance. By employing cross-layer feature reuse and parallel attention mechanisms, it adaptively suppresses redundant background in blurred feature maps while enhancing spatial positional weights highly correlated with change boundaries. The structure of CLCR is illustrated in Fig. 6.

Let $\mathbf{D}_n$ denote the initial differential feature at the current layer, and $\hat{\mathrm{D}}_{n-1}$ denote the enhanced differential feature from the preceding layer. To achieve lossless information reuse, we inject the semantic context from the preceding layer into the current layer via pixel-wise addition, as follows:

$$\mathbf{D}'_n = \text{Conv}_{3\times 3}\left(\mathbf{D}_n + \mathcal{U}_{\text{bilinear}}\left(\hat{\mathrm{D}}_{n-1}\right)\right). \tag{23}$$

Subsequently, to extract true boundary information from $\mathbf{D}'_n$ and suppress pseudo-change noise caused by environmental differences (e.g., illumination and seasonality), we design a parallel spatial-channel attention mechanism (CSSA). This mechanism computes saliency weights in both spatial and channel domains, thereby assigning higher response values to truly changed regions. The spatial-channel weights are formulated in the following equation:

$$\begin{cases} \boldsymbol{W}_{\text{SSA}} &= \sigma\left(\text{Conv}^{c\to 1}_{1\times 1}\left(\mathbf{D}'_n\right)\right) \\ \boldsymbol{W}_{\text{CSA}} &= \sigma\left(\text{MLP}\left(\text{GAP}\left(\mathbf{D}'_n\right)\right)\right) \end{cases} \tag{24}$$

where $\text{MLP}(\cdot)$ represents the fully connected layers, $\text{GAP}(\cdot)$ denotes the global average pooling, $\boldsymbol{W}_{\text{SSA}}$ denotes the spatial weight, and $\boldsymbol{W}_{\text{CSA}}$ denotes the channel weight. The weighted features are obtained via pixel-wise multiplication, as follows:

$$\begin{cases} \mathbf{D}_{\text{SSA}} &= \mathbf{D}'_n \otimes \boldsymbol{W}_{\text{SSA}} \\ \mathbf{D}_{\text{CSA}} &= \mathbf{D}'_n \otimes \boldsymbol{W}_{\text{CSA}}. \end{cases} \tag{25}$$

Finally, the MSDConv module is employed to flexibly capture boundary representations at different scales by leveraging its cyclic multiscale dilated convolution strategy and

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spatial–spectral fusion cooperative attention strategy [51], thereby effectively enhancing the model's perception capability of differential boundaries, as follows:

$$\hat{\mathbf{D}}_n = \Psi_{\mathrm{MSD}}\left(\mathbf{D}_{\mathrm{SSA}} + \mathbf{D}_{\mathrm{CSA}}\right) \tag{26}$$

where $\mathbf{D}_{\mathrm{SSA}}$ denotes the spatial attention weighted output, and $\mathbf{D}_{\mathrm{CSA}}$ denotes the channel attention weighted output.

E. Loss Function

DDPM conducts self-supervised training by measuring the discrepancy between the predicted noise and the actual noise. Given that the noise follows a Gaussian distribution, the mse loss function is employed to train the DDPM by minimizing the discrepancy between the predicted and actual noise, as formulated in (27). This selection is motivated by the effectiveness of mse in estimating the parameters of Gaussian noise distributions, thereby rendering the DDPM denoising process more precise

$$\mathcal{L}_{\mathrm{MSE}} = \left(\boldsymbol{\epsilon} - \boldsymbol{\epsilon}_\theta\left(\mathbf{x}_t, t\right)\right)^2 \tag{27}$$

where $\boldsymbol{\epsilon}$ represents the added actual noise, $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ denotes the noise predicted by the model, $\mathbf{x}_t$ represents the noisy image, and t denotes the time step.

In RSCD tasks, changed regions typically constitute a minimal proportion, and this is a pixel-level binary classification problem. Given that DDPM generates features with smooth semantic continuity, to better adapt these diffusion features to the requirements of the RSCD task, we employ the cross-entropy (CE) loss function, as formulated in (28). This loss effectively measures the discrepancy between model predictions and ground-truth (GT) labels, thereby optimizing the overall performance of the CD network

$$\mathcal{L}_{\mathrm{CE}} = -\frac{1}{n}\sum_{i=1}^{n}\left[\mathbf{Y}_i \log\left(\hat{\mathrm{Y}}_i\right) + \left(1 - \mathbf{Y}_i\right)\log\left(1 - \hat{\mathrm{Y}}_i\right)\right] \tag{28}$$

where $\mathbf{Y}_i$ denotes the GT label, and $\hat{\mathrm{Y}}_i$ represents the model output.

IV. Experiment and Results

A. Datasets

To validate the effectiveness of S2S-MGPNet, we conducted extensive experiments on three public CD datasets: WHU-CD [52], CDD-CD [53], and LEVIR-CD [54].

WHU-CD is a public dataset for building CD in RS images. It possesses a spatial dimension of $32\,507 \times 15\,354$ pixels with a spatial resolution of 0.2 m. Since this dataset lacks a unified segmentation rule, we cropped the images into nonoverlapping patches of size 256×256.

CDD-CD comprises 11 image pairs featuring seasonal variations. Beyond building changes, it incorporates diverse change categories including vehicles, vegetation, and roads. Specifically, it consists of seven pairs with dimensions of 4725×2700 pixels and four pairs of 1900×1000 pixels, with spatial resolutions ranging from 0.03 to 1 m. The images were cropped into patches of size 256×256 and randomly partitioned into 10 000, 3000, and 3000 samples for training, validation, and testing, respectively.

LEVIR-CD is a large-scale public dataset for building CD, comprising 637 pairs of high-resolution RS images. Each image has a size of 1024×1024 pixels and a spatial resolution of 0.5 m. We cropped these images into patches of size 256×256, resulting in 7120, 1024, and 2048 bitemporal image pairs for training, validation, and testing, respectively.

B. Implementation Details and Evaluation Metrics

The proposed S2S-MGPNet network was implemented using PyTorch 2.2 and trained for 120 epochs on a single NVIDIA GeForce RTX 3090 GPU with 24 GB of VRAM. We employed the AdamW optimizer with a momentum of 0.99. The initial learning rate was set to 1×10^{-4} and decayed linearly during training with a decay factor of 0.01. To quantitatively evaluate the proposed method, we adopt five standard metrics: $F1$-score and IoU serve as the primary indicators, supplemented by precision (Pre), recall (Rec), and overall accuracy (OA). The definitions of these metrics are as follows:

$$Pre = \frac{TP}{TP + FP} \tag{29}$$

$$Rec = \frac{TP}{TP + FN} \tag{30}$$

$$IoU = \frac{TP}{TP + FN + FP} \tag{31}$$

$$F1 = \frac{2 \times Pre \times Rec}{Pre + Rec} \tag{32}$$

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{33}$$

where TP, FP, FN, and TN denote true positive (TP), false positive (FP), false negative (FN), and true negative (TN), respectively. $F1$ represents the combination of Pre and Rec, while IoU and OA are calculated based on the confusion matrix.

C. Comparison With State-of-the-Art Methods

To validate the superiority of S2S-MGPNet, we compare it with several state-of-the-art RSCD methods, which can be broadly categorized into three groups. CNN-based methods: DMINet [23], CCLNet [55], USSFCNet [51], SEIFNet [56], mAFGNet [44]. Transformer-based methods: BIT [25], ScratchFormer [57], M-Swin [58], STRobustNet [59], ConvFormer [60]. DDPM-based methods: DDPM-CD [16], and GCD-DDPM [40].

1) *Quantitative Evaluation:* We conducted extensive experiments on three large-scale public datasets, with the results presented in Table I. Our method consistently outperforms other CNN-, Transformer-, and DDPM-based approaches across all three datasets. On the WHU-CD dataset, experimental results demonstrate that our method achieves an IoU improvement of 2.05% over the state-of-the-art CNN-based mAFGNet, 2.57% over the Transformer-based M-Swin, and 2.55% over the DDPM-based DDPM-CD. On the CDD dataset, our approach surpasses the leading CNN-based USSFCNet by 4.25% in IoU, outperforms the Transformer-based ScratchFormer by 0.47%, and exceeds the DDPM-based

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TABLE I

QUANTITATIVE COMPARISONS OF OA, IOU, $F1$, REC, AND PRE ON THREE RSCD DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	Ref.	WHU-CD					CDD-CD					LEVIR-CD				
		OA	IoU	F1	Rec	Pre	OA	IoU	F1	Rec	Pre	OA	IoU	F1	Rec	Pre
CNN-based Methods																
DMINet [23]	TGRS 23	98.97	79.68	88.69	86.25	93.84	98.43	89.32	93.89	92.24	94.93	99.07	82.99	90.71	89.95	92.52
CCLNet [55]	TGRS 23	99.00	78.54	87.98	89.36	86.64	98.53	89.65	94.24	93.47	95.86	99.04	82.00	90.11	89.47	90.76
USSFCNet [51]	TGRS 23	99.38	85.38	92.11	90.78	93.49	98.63	89.86	94.66	93.85	95.48	99.04	82.55	90.44	89.11	91.81
SEIFNet [56]	TGRS 24	98.90	76.04	86.39	85.77	87.01	98.55	89.76	94.27	93.78	95.97	99.09	83.40	90.95	89.46	92.49
mAFGNet [44]	TGRS 24	**99.41**	**87.02**	**93.06**	91.74	94.42	98.87	89.24	94.31	94.23	94.40	99.12	**84.32**	91.30	90.08	**92.59**
Transformer-based Methods																
BIT [25]	TGRS 22	99.33	86.35	92.67	92.08	93.28	99.10	79.30	88.51	90.75	86.38	99.09	83.57	91.05	90.55	91.55
ScratchFormer [57]	TGRS 24	99.32	84.97	91.87	89.78	92.44	**99.29**	**93.94**	**96.58**	**97.12**	**98.43**	99.14	84.28	91.40	90.51	92.33
M-Swin [58]	TGRS 24	99.36	86.09	92.93	91.37	94.55	99.10	92.44	95.68	94.42	95.60	99.05	83.57	91.02	90.05	92.07
STRobustNet [59]	TGRS 25	99.32	83.29	90.89	88.94	92.92	97.83	92.44	95.70	94.65	95.77	99.10	83.66	91.11	90.67	91.54
ConvFormer [60]	TGRS 25	99.27	85.54	92.13	89.49	**95.11**	98.94	91.53	95.57	95.84	95.31	99.13	84.23	**91.44**	90.52	92.37
DDPM-based Methods																
DDPM-CD [16]	WACV 25	99.38	86.31	92.65	92.06	93.23	98.98	91.62	95.62	94.43	95.05	99.11	83.40	91.02	89.38	**92.93**
GCD-DDPM [40]	TGRS 24	99.39	86.52	92.54	**92.29**	92.79	98.87	90.05	94.93	95.10	94.76	**99.14**	83.56	90.96	**91.24**	90.68
S2S-MGPNet (Ours)	–	**99.53**	**89.02**	**94.20**	**93.90**	**94.49**	**99.33**	**94.47**	**97.16**	**96.98**	**97.15**	**99.15**	**84.48**	**91.58**	**90.78**	92.40

DDPM-CD by 2.79%. On the LEVIR-CD dataset, our method yields IoU gains of 0.12%, 0.06%, and 0.94% compared to the state-of-the-art CNN-based mAFGNet, Transformer-based ConvFormer, and DDPM-based GCD-DDPM, respectively.

2) *Qualitative Evaluation:* The visualization results of different methods on the three RSCD datasets are presented in Fig. 7. To enhance visualization clarity, we employ a four-color coding scheme to represent the detection results: TP is shown in white, FP in red, TN in black, and FN in blue. The results demonstrate that S2S-MGPNet exhibits superior performance in the following aspects.

1) Positioning Accuracy Advantage: S2S-MGPNet demonstrates significant advantages in change object localization. Compared to existing methods, our approach not only identifies change regions more accurately but also effectively mitigates the inherent stochasticity of DDPM, thereby providing more precise localization of RS semantic features. Specifically, as illustrated in the WHU-CD samples in Fig. 7 (row 1), methods such as USSFCNet, BIT, ScratchFormer, and M-Swin struggle to capture building semantic information, leading to incomplete prediction results. Meanwhile, although DDPM-CD and GCD-DDPM are based on diffusion models, they fail to suppress the stochasticity of the diffusion process, resulting in blurred building boundaries and compromised detection accuracy. In the CDD-CD dataset, small-target semantic details exist (Fig. 7, CDD-CD: row 1, semantic changes in the four flower beds); GCD-DDPM tends to overlook these fine-grained details, resulting in missed detections. mAFGNet and DDPM-CD struggle to accurately localize these small targets. In contrast, S2S-MGPNet accurately and comprehensively identifies these minute change regions, achieving precise semantic localization and demonstrating outstanding small object detection capabilities.

2) Superior Noise Resistance Advantages: The CDD-CD dataset contains RS images with poor visual quality and severe noise interference (Fig. 7, CDD-CD: rows 2, 3). The noise causes significant blurring of relevant semantic information. Many existing advanced methods, such as USSFCNet, mAFGNet, M-Swin, and DDPM-CD, struggle to effectively extract semantic information from these noise-contaminated images, leading to both missed detections and false alarms in true change regions. In contrast, S2S-MGPNet fully leverages the robust noise adaptability of DDPM to accurately extract key semantic features from blurred images. It effectively suppresses the inherent stochasticity of DDPM via an MCFA strategy, while simultaneously balancing global distributions and local details through neighborhood feature aggregation. This innovative strategy enables precise feature extraction even on highly noisy images, achieving accurate recognition of RS semantic features.

3) Advantages in Discriminating Pseudo-Changes: In Fig. 7, WHU-CD, second row, building changes and road color variations share a similar white appearance, making them highly prone to confusion. Many existing methods, such as USSFCNet, ScratchFormer, DDPM-CD, and GCD-DDPM, exhibit extensive red regions in the CD maps, indicating high false detection rates and an inability to effectively eliminate these road pseudo-changes. In contrast, S2S-MGPNet accurately identifies building boundaries and bodies, significantly mitigating the interference of road pseudo-changes. Furthermore, seasonal factors cause variations in lawn color within the CDD-CD dataset (Fig. 7, CDD-CD: row 1). Methods like mAFGNet, BIT, ScratchFormer, and GCD-DDPM fail to effectively exclude these seasonal variations, resulting in noticeable false detections. However, S2S-MGPNet effectively handles lawn pseudo-changes induced by seasonality, further validating its outstanding capability in suppressing pseudo-changes within complex RS scenarios. Collectively, these experimental results demonstrate the significant advantages of the proposed method in distinguishing genuine semantic changes from various types of pseudo-changes.

4) Boundary Integrity Advantages: S2S-MGPNet exhibits exceptional performance in preserving the integrity of change boundaries. This is primarily attributed to our proposed CLCR,

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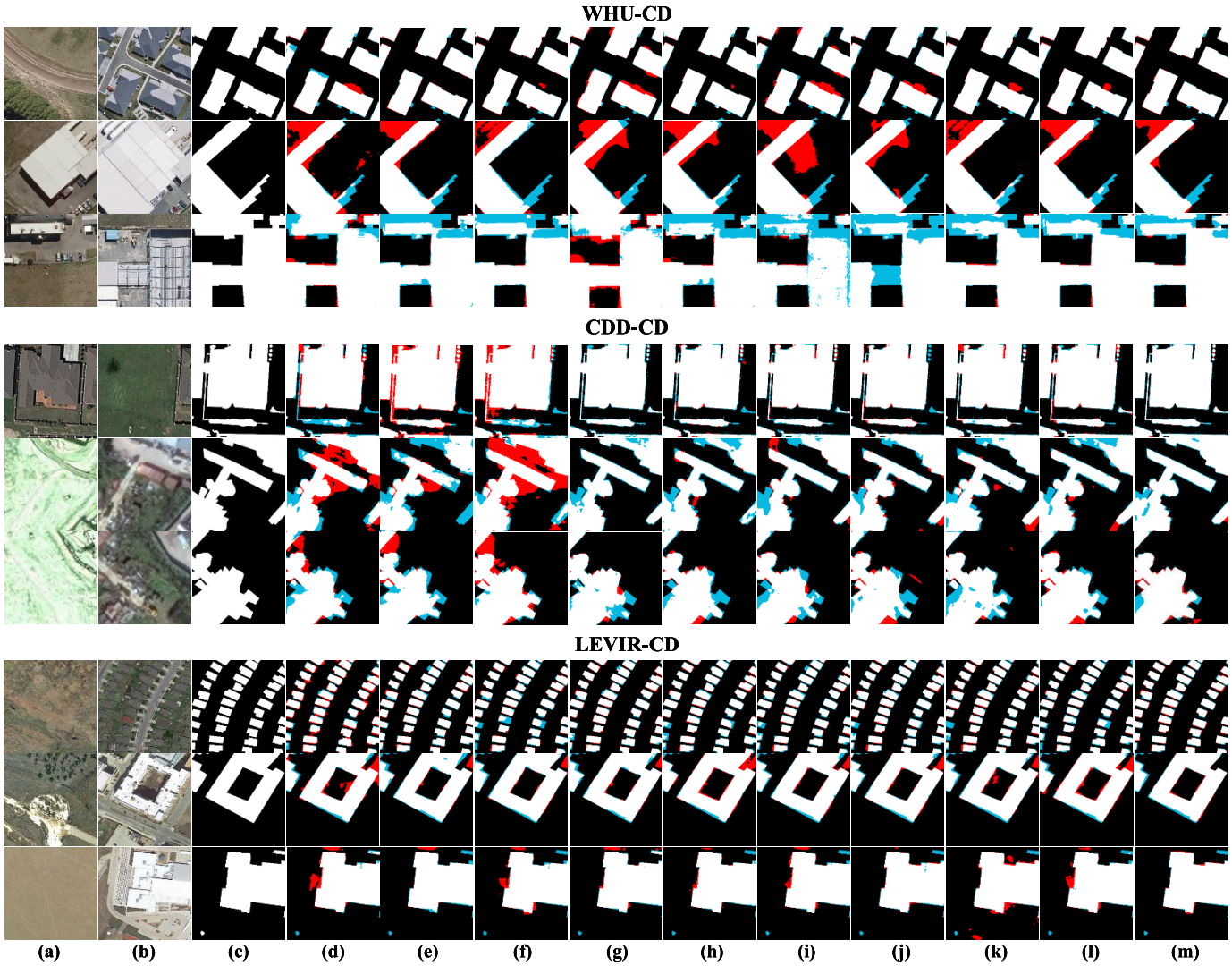


Fig. 7. Visual comparisons of results on the three datasets. (a) Image T1. (b) Image T2. (c) GT. (d) USSFCNet. (e) mAFGNet. (f) BIT. (g) ScratchFormer. (h) M-Swin. (i) STRobustNet. (j) ConvFormer. (k) DDPM-CD. (l) GCD-DDPM. (m) S2S-MGPNet (Ours). White, black, red, and blue denote TP, TN, FP, and FN, respectively.

which eliminates interference from redundant information while reusing and enhancing differential boundary information across various structural scales, thereby achieving high-quality boundary reconstruction. In the LEVIR-CD dataset, there are changes in large-scale building clusters (Fig. 7, LEVIR-CD: row 1); S2S-MGPNet accurately delineates the boundaries of each individual building. Furthermore, when building boundaries are obscured by shadows (Fig. 7, LEVIR-CD: row 2), traditional DDPM-based CD methods struggle to extract complete boundaries. By processing distributed differential features, S2S-MGPNet yields more complete building boundaries.

TABLE II

EXPERIMENTAL RESULTS OF DIFFERENT TIME STEP VALUES. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Time Steps (T_n)	WHU-CD			CDD-CD			LEVIR-CD		
	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU
$T_n = 50$	99.48	93.49	88.04	99.16	96.45	93.36	99.08	90.91	83.34
$T_n = 150$	99.47	93.26	87.38	99.04	95.92	92.17	99.08	90.79	83.14
$T_n = 250$	99.45	93.05	87.00	98.99	95.70	91.75	99.01	90.14	82.32
$T_n = \{25, 50, 75\}$	99.49	93.62	88.14	99.21	96.66	93.54	99.10	91.11	83.60
$T_n = \{50, 150, 250\}$	**99.53**	**94.20**	**89.02**	**99.32**	**97.11**	**94.41**	**99.15**	**91.58**	**84.48**
$T_n = \{50, 100, 400\}$	**99.51**	**93.80**	**88.76**	**99.22**	**96.70**	**93.61**	**99.11**	**91.12**	**83.68**

D. Ablation Analysis

1) Time Step Analysis: Determining the optimal time step value t and the number of time steps n is critical for balancing noise robustness, semantic preservation, and computational overhead. A larger t enhances the model's robustness against strong noise but risks degrading the original semantic information; conversely, a smaller t better preserves fine-grained details but limits noise adaptability. Consequently, the combination of three time steps can effectively cover the low, medium, and high noise intensity ranges. Specifically, the low-noise stage retains high-frequency image details, the medium-noise stage balances semantics and textures, and the high-noise stage extracts abstract semantic information, thereby enhancing the model's noise robustness. Alternative

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TABLE III

ABLATION STUDY OF S2S-MGPNET ON THREE DATASETS. ABLATION ANALYSIS OF MCST AND CLCR. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

No.	MCST	CLCR	WHU-CD			CDD-CD			LEVIR-CD		
			OA	F1	IoU	OA	F1	IoU	OA	F1	IoU
1			99.42	92.65	86.31	98.98	95.62	91.62	99.07	90.72	83.40
2	✓		**99.49**	**93.82**	**88.32**	**99.21**	**96.64**	**93.51**	99.09	90.98	83.45
3		✓	99.47	93.45	87.72	99.06	96.01	92.32	**99.12**	**91.11**	**83.61**
4	✓	✓	**99.53**	**94.20**	**89.02**	**99.32**	**97.11**	**94.41**	**99.15**	**91.58**	**84.48**

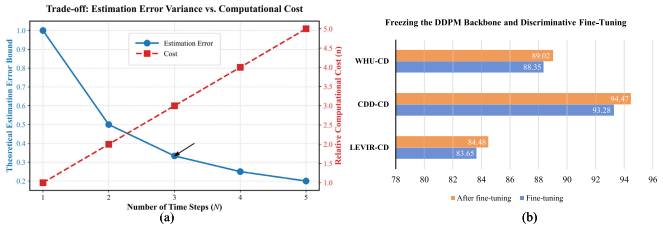


Fig. 8. Parameter trade-off and backbone performance. (a) Trade-off analysis between noise error and computational cost. The blue solid line and red dashed line represent the image noise error versus the number of time steps N, and the relative computational cost ratio, respectively. The optimal balance point for the MCFA is at $N = 3$. (b) Performance comparison between the frozen (orange) and partially fine-tuned (blue) diffusion backbones.

combinations exhibit inferior noise stratification, resulting in lower stability after feature aggregation compared to our proposed combination. Furthermore, while increasing the number of steps n facilitates the aggregation of richer information, it inevitably incurs higher computational costs. Experimental results for different time step configurations are presented in Table II, and the trade-off between noise variance suppression and computational overhead with varying n is illustrated in Fig. 8(a).

The results demonstrate that the multi-time-step strategy significantly outperforms single-time-step configurations (Rows 1–3). As indicated in Rows 4 and 6, excessively weak or strong noise levels lead to insufficient noise adaptability or the disruption of original semantic distributions, respectively. As illustrated in Fig. 8(a), when the number of steps n exceeds 3, the marginal performance improvement does not justify the significant increase in computational cost. Consequently, given that the RSCD task relies on both fine-grained structural change recognition and global semantic consistency modeling, we select representative time steps from the early, middle, and late stages of the diffusion process: low noise ($t = 50$, preserving high-frequency details), medium noise ($t = 150$, balancing semantics and texture), and high noise ($t = 250$, ensuring noise robustness). This strategy achieves a complementary fusion of multiscale semantic information while balancing computational efficiency.

2) Impact of Freezing the Generative Prior: To validate the specific role of the "universal generative prior" learned by the pretrained DDPM, we compared the completely frozen backbone with a fine-tuned backbone, in which the last two layers of the DDPM decoder were unfrozen during CD training. The results shown in Fig. 8(b), discriminative supervised fine-tuning of the DDPM backbone leads to performance degradation. Furthermore, we replaced the CLCR strategy with

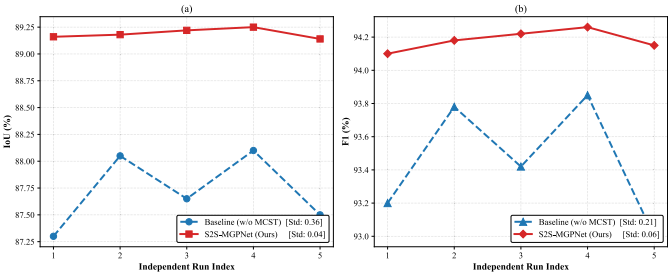


Fig. 9. Quantitative comparison of statistical stability across five independent runs on the WHU-CD dataset. (a) Fluctuation of IoU. (b) Fluctuation of $F1$. The solid red line indicates that the proposed S2S-MGPNet effectively suppresses the severe performance fluctuations observed in the baseline network.

traditional convolution for decoding (see Table III, row 2). On unseen test sets, our model maintains a higher IoU than SOTA DDPM-based CD methods even after removing the CLCR strategy. These results indicate that the frozen DDPM provides an optimal feature manifold. The CLCR strategy serves as an auxiliary, while discriminative fine-tuning disrupts the established "universal generative prior."

3) MCST Effectiveness: MCST aims to mitigate the inherent stochasticity of DDPM features. To validate its efficacy, we integrated MCST into the model and compared it against a baseline model employing solely standard convolution operations. The quantitative comparisons are presented in the second row of Table III. The results indicate that the incorporation of MCST yields significant performance gains over the baseline across all three datasets. Specifically, on the WHU-CD dataset, the $F1$-score and IoU improved by 1.17% and 2.01%, respectively. To verify the "Stochasticity to Stability" transition, we conducted a statistical variance analysis to quantify the suppression of result fluctuations. We performed five independent and complete inferences on three test sets using different random noise. The Std of these runs was calculated, with the results presented in Table IV. For an intuitive comparison, Fig. 9 visualizes the fluctuations on the WHU dataset via a line chart. Removing the MCST strategy leads to significant performance fluctuations. In contrast, the proposed S2S-MGPNet achieves an extremely low standard deviation. To verify the stability of the features, we computed the difference between the MCST-enhanced features and visualized the resulting differential features using the t-SNE algorithm. As illustrated in Fig. 10, the enhanced features exhibit more distinct interclass boundaries and compact intraclass distributions. These results substantiate that by suppressing the stochasticity of the diffusion process and incorporating multiscale spatial complementary information, MCST effectively enhances the representational capability of distributed diffusion features, thereby achieving more precise CD.

4) CLCR Effectiveness: In our proposed S2S-MGPNet, the CLCR is employed to progressively reinforce differential information across feature layers. As evidenced in Table III (Row 3), substituting the baseline's 3×3 convolutions with CLCR yields performance gains of 0.8% in $F1$-score and 1.41% in IoU on the WHU-CD dataset. The heatmaps in Fig. 11 further visually confirm that CLCR effectively

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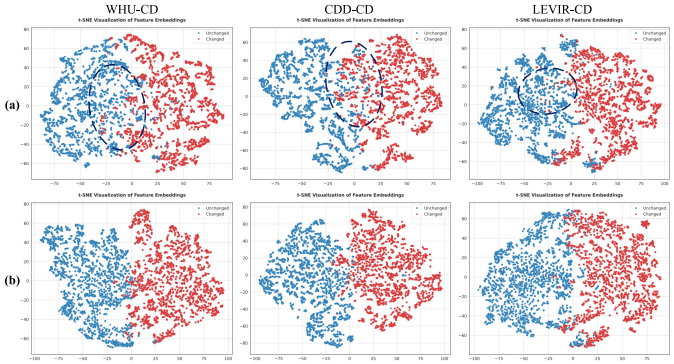


Fig. 10. Illustration of MCST effectiveness using t-SNE visualization of internal differential features in S2S-MGPNet. Red and blue denote changed and unchanged features, respectively. (a) Baseline network. (b) Feature enhancement with MCST.

TABLE IV
STABILITY ANALYSIS OF THE PROPOSED MCST STRATEGY OVER FIVE INDEPENDENT RUNS

Dataset	Method	IoU (%)	F1-Score (%)
WHU-CD	w/o MCST	87.72 ± 0.36	93.45 ± 0.21
	Ours	$\mathbf{89.19 \pm 0.04}$	$\mathbf{94.18 \pm 0.06}$
CDD	w/o MCST	92.06 ± 0.09	95.87 ± 0.08
	Ours	$\mathbf{94.46 \pm 0.01}$	$\mathbf{97.15 \pm 0.01}$
LEVIR-CD	w/o MCST	83.67 ± 0.25	91.11 ± 0.20
	Ours	$\mathbf{84.48 \pm 0.02}$	$\mathbf{91.58 \pm 0.04}$

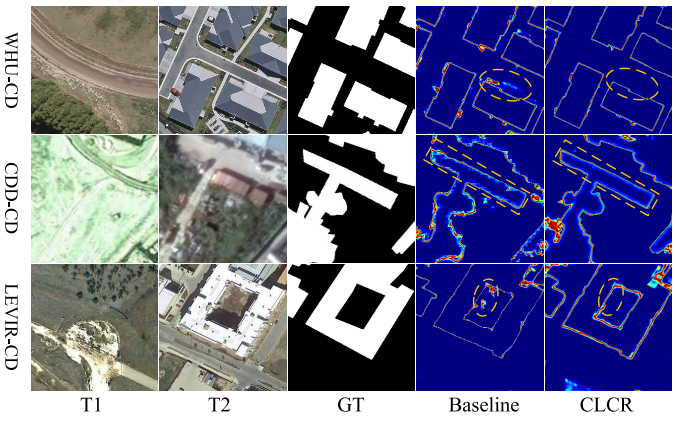


Fig. 11. Heatmaps validating the effectiveness of CLCR. T1, T2, and GT denote the input bitemporal images and GT, respectively. Baseline represents prediction results using solely convolutional upsampling, while CLCR denotes prediction results enhanced by the CLCR strategy.

sharpens fuzzy differential boundaries. To analyze the contributions of CLCR's internal mechanisms, we conducted component-level ablation studies (Table V). First, removing the CSSA strategy leads to decreases of 0.25% in $F1$ and 0.42% in IoU (Row 2 versus Row 4), indicating that CSSA effectively filters redundant features. Second, replacing the MSDConv block with a conventional 1×1 convolution while retaining CSSA results in performance drops of 0.35% and 0.72% (Row 3), validating the role of MSDConv in boundary refinement. The synergy between these two

TABLE V
ABLATION STUDY OF S2S-MGPNET ON THREE DATASETS. DETAILED ABLATION ANALYSIS OF INTERNAL SUBMODULES IN MCST AND CLCR (NSFS, CSSA, AND MSDCONV). THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

No.	Sub-Modules			WHU-CD			CDD-CD			LEVIR-CD		
	NSFS	CSSA	MSDConv	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU
1		✓	✓	99.47	93.45	87.72	99.06	96.01	92.32	99.12	91.11	83.61
2	✓		✓	**99.52**	**93.95**	**88.60**	**99.21**	**96.68**	**93.62**	**99.13**	**91.37**	**84.21**
3	✓	✓		99.50	93.85	88.30	**99.21**	96.66	93.57	99.12	91.27	83.95
4	✓	✓	✓	**99.53**	**94.20**	**89.02**	**99.32**	**97.11**	**94.41**	**99.15**	**91.58**	**84.48**

TABLE VI
ABLATION STUDY OF S2S-MGPNET ON THREE DATASETS. COMPARATIVE VALIDATION OF MULTISCALE FEATURE FUSION STRATEGIES. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

Method	Ref.	WHU-CD			CDD-CD			LEVIR-CD		
		OA	F1	IoU	OA	F1	IoU	OA	F1	IoU
Baseline	–	99.47	93.45	87.72	98.98	95.62	91.62	99.11	91.02	83.40
MSGDC [61]	CVPR 25	**99.50**	**93.66**	**88.08**	99.02	95.84	92.02	**99.11**	**91.26**	**83.93**
HFF [62]	BSPC 24	99.48	93.44	87.69	**99.29**	**96.82**	**94.12**	99.09	90.98	83.46
MAF [44]	TGRS 24	99.49	93.62	88.00	98.97	95.62	91.62	99.10	91.06	83.59
NSFS (Ours)	–	**99.53**	**94.20**	**89.02**	**99.32**	**97.11**	**94.41**	**99.15**	**91.58**	**84.48**

components significantly empowers CLCR to capture and refine differential information, thereby optimizing the overall model performance.

5) Discussion on the NSFS Strategy: To validate the effectiveness of NSFS within MCST for reinforcing distributed diffusion features, further ablation experiments were conducted. First, removing the NSFS strategy while relying solely on the MCFA strategy (Table V, Row 1) resulted in $F1$ and IoU decreases of 0.75% and 1.3% on the WHU-CD dataset, confirming that although the inherent skip connections in the DDPM facilitate multiscale feature fusion, their primary goal is to reconstruct generative features for noise prediction rather than optimizing discriminative representations for downstream CD tasks. Consequently, despite these internal connections, the extracted single-layer features still exhibit representation ambiguity. Subsequently, to demonstrate the superiority of NSFS as a calibration mechanism tailored for discriminative CD tasks, we benchmarked it against several advanced multiscale fusion strategies, including the dilated convolution-based MSGDC [61], the adaptive attention-based HFF [62], and the feature pyramid-based MAF [44]. The results in Table VI demonstrate that NSFS significantly outperforms these general-purpose strategies, proving its superior adaptability to distributed diffusion features with pronounced spatial hierarchical correlations.

E. Analysis of Complexity, Distribution Shift and Limitations

1) Computational Complexity: Table VII illustrates the trade-off between performance and efficiency between our proposed S2S-MGPNet and other CD methods. Although S2S-MGPNet exhibits higher computational complexity than lightweight CNNs, it reduces computational complexity by 79% compared to existing large-scale Transformer-based methods (ScratchFormer). Furthermore, compared to SOTA DDPM-based CD methods (GCD-DDPM), S2S-MGPNet achieves a 59% reduction in parameter count and an 84% reduction in computational complexity. Crucially, our method

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TABLE VII
COMPARISON OF COMPUTATIONAL COMPLEXITY AND PERFORMANCE ON THREE DATASETS

Method	IoU (%)			Params (M)	FLOPs (G)
	WHU-CD	CDD-CD	LEVIR-CD		
DMINet [23]	79.68	89.32	82.99	6.24	14.42
CCLNet [55]	78.54	89.65	82.00	28.78	46.54
USSFCNet [51]	85.38	89.86	82.55	**1.52**	**4.86**
SEIFNet [56]	76.04	89.76	83.40	27.91	8.37
mAFGNet [44]	87.02	89.24	84.32	6.58	62.18
BIT [25]	86.35	79.30	83.57	17.19	15.17
ScratchFormer [57]	84.97	93.94	84.28	36.95	198.39
STRobustNet [59]	83.29	92.44	83.66	5.23	13.19
ConvFormer [60]	85.54	91.53	84.23	150.66	20.39
DDPM-CD [16]	86.31	91.62	83.40	46.41	32.84
GCD-DDPM [40]	86.52	90.05	83.56	130.83	269.52
S2S-MGPNet (Ours)	**89.02**	**94.47**	**84.48**	53.07	41.80

achieves superior detection accuracy at a manageable computational cost, with IoU scores significantly outperforming competing methods.

2) Discussion on Distribution Shift: A potential concern is the domain gap between the global pretraining data and the benchmark datasets such as WHU or CDD. However, we argue that the DDPM extracts fundamental semantic priors, such as object geometry, which are inherently transferable. The inherent distribution shift is effectively mitigated by our two-stage paradigm: while the DDPM provides a stable feature foundation, the proposed MCST and CLCR strategies act as learnable adaptation layers to refine these features using task-specific labels. Furthermore, the CD task emphasizes modeling the differences between bi-temporal images of the same region, and this relative change pattern provides robustness against distribution shifts. The performance improvements across three different datasets (as shown in Table I) verify that the advantages of large-scale generative priors in suppressing noise and capturing complex geometries significantly outweigh the challenges posed by domain shifts.

3) Limitations and Potential Improvements: While S2S-MGPNet achieves accurate CD on high-resolution images, the pretrained DDPM's network layers are too deep, resulting in extracted features with high dimensions and channel counts. Consequently, enhancing these features inevitably incurs considerable parameter overhead and computational costs. To alleviate computational bottlenecks and reduce pretraining costs, future work can explore the following optimizations: First, latent diffusion architectures can be adopted to shift feature extraction from the pixel space to a latent space by utilizing a VAE to compress the images. Second, model pruning techniques can be employed to develop a lightweight DDPM, thereby reducing the overall model complexity. Finally, a lightweight discriminative model could be introduced to provide deterministic representations. This would reduce the required number of independent sampling steps in the DDPM, effectively lowering computational complexity while preserving its powerful generative semantic representation capabilities.

V. CONCLUSION

This article proposes a Monte Carlo-guided generative prior paradigm based on a pretrained DDPM, establishing a novel semantic representation paradigm for RSCD. To address the inherent stochasticity and local representation deficiencies of pretrained DDPM, we designed the MCST strategy. By employing a Monte Carlo feature ensemble and a spatial multilayer fusion strategy, MCST effectively mitigates feature stochasticity while enhancing multiscale representations. Furthermore, the proposed CLCR strategy precisely captures fine-grained structures in change regions via coarse-to-fine differential information aggregation, effectively resolving the issue of blurred boundaries in diffusion features. This work not only fully exploits the potential of DDPM in RSCD but also provides a robust and scalable baseline for DDPM-assisted RSCD tasks.

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